Fashion Recommendation System

Personalized Fashion Product Recommendations Using Collaborative Filter

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Phase 4 Project – Advanced Machine Learning

Business Problem

Online fashion platforms often struggle to:

- Help users discover products they may like
- Keep users engaged on the platform
- Provide personalized shopping experiences
- Reduce decision fatigue caused by too many product options

Without personalization, customers may lose interest and leave the platform without making purchases.

Project Objective

The goal of this project was to build a recommendation system that:

- Suggests fashion products to users
- Uses user rating behavior to identify similar preferences
- Improves personalization within fashion e-commerce platforms
- Demonstrates how machine learning can improve customer experience

Dataset Overview

The dataset included:

- User IDs
- Product IDs
- Product names
- Fashion categories
- Brands
- Ratings
- Prices

Categories included:

- Men's Fashion
- Women's Fashion
- Kids' Fashion

Brands included:

- Nike
- Adidas
- Gucci
- Zara
- H&M

Project Workflow

The project followed these major steps:

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Recommendation Modeling
5. Recommendation Improvement
6. Dashboard Visualization
7. Streamlit Application Development

Recommendation System Approach

This project used Collaborative Filtering

The model recommends products based on:

- Similar users
- Similar rating patterns
- Shared preferences

If users liked similar products in the past, the system assumes they may enjoy similar products in the future.

Improving Recommendation Quality

To improve recommendation robustness:

- A popularity-based fallback system was added
- Highly rated products were identified across all users
- This helped support users with limited interaction history

This improved the reliability of recommendations.

Tableau Dashboard

An interactive Tableau dashboard was created to visualize:

- Distribution of products by category
- Average ratings across brands
- Product rating distributions

The dashboard helps stakeholders understand:

- Product trends
- Customer preferences
- Brand performance

Streamlit Web Application

A simple interactive web application was developed where:

- Users enter a User ID
- The system generates personalized recommendations
- Recommended products are displayed instantly

The application demonstrates how the recommendation system works in practice.

Key Insights

Users with similar behaviors tend to prefer similar products

Recommendation systems can improve personalization

Popularity-based fallback methods improve recommendation stability

Fashion brands showed slight differences in average ratings

Challenges Encountered

Some challenges included:

- Library compatibility issues
- Streamlit dependency conflicts
- Recommendation quality improvement
- Preparing clean and interpretable outputs

These were resolved through debugging, package management, and model refinement.

Recommendations

Future improvements could include:

- Hybrid recommendation systems
- Real-time recommendation updates
- User authentication and profiles
- Product image integration
- Deep learning recommendation models

Conclusion

This project successfully demonstrated how machine learning can:

- Personalize fashion recommendations
- Improve user experience
- Support e-commerce decision-making

The recommendation system, dashboard, and web application together provide a complete demonstration of an end-to-end recommendation solution.

Thank You

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